

Phase-coupled models for synchronization with nonlocal temporal interactions

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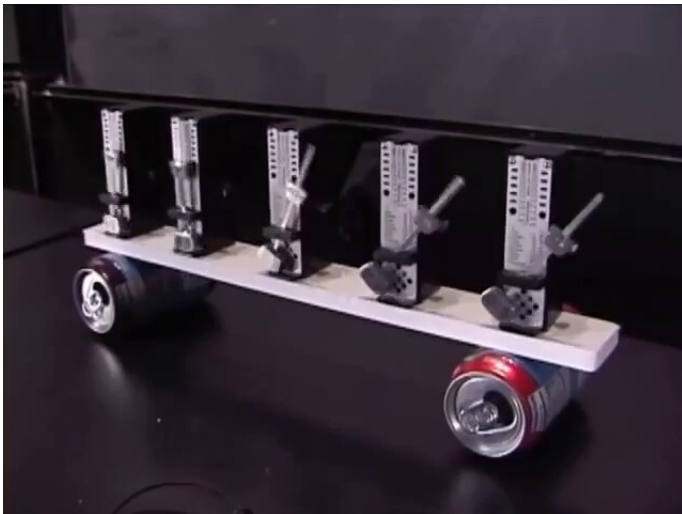
Kuramoto model for synchronization

Kuramoto model with memory

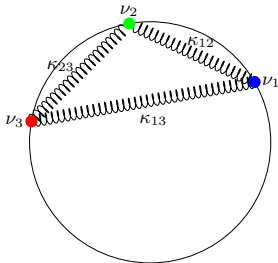
Synchronization under the effect of memory

Kuramoto model for synchronization

Synchronization



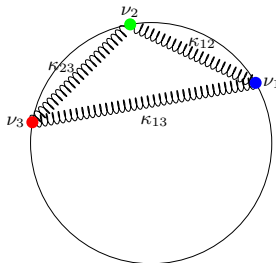
Kuramoto model



- (Newton's law) $F = ma$
 $\implies m\ddot{\theta}_i = -\gamma\dot{\theta}_i + \nu_i + (\text{elastic torque on } i\text{-th oscillator}).$
- (elastic energy between i -th and j -th oscillator)

$$U_{ij} = \kappa_{ij}(1 - \cos(\theta_i - \theta_j)) \implies -\frac{\partial U_{ij}}{\partial \theta_i} = \kappa_{ij} \sin(\theta_j - \theta_i).$$

Kuramoto model



- Inertial Kuramoto model:

$$m\ddot{\theta}_i = -\gamma\dot{\theta}_i + \nu_i + \sum_{j=1}^N \kappa_{ij} \sin(\theta_j - \theta_i)$$

- Small inertia/high friction regime:

$$\frac{m}{\gamma} \approx 0 \implies \dot{\theta}_i = \nu_i + \sum_{j=1}^N \kappa_{ij} \sin(\theta_j - \theta_i) \quad [\text{Kuramoto, 1975}]$$

Kuramoto model

$$\dot{\theta}_i = \nu_i + \frac{\kappa}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N$$

- θ_i : phase of i -th oscillator on unit circle
- ν_i : natural frequency of i -th oscillator
- κ : nonnegative coupling strength
- $\exists \lim_{t \rightarrow \infty} |\theta_i(t) - \theta_j(t)| \implies \lim_{t \rightarrow \infty} |\dot{\theta}_i(t) - \dot{\theta}_j(t)| = 0$
 (phase-locking) (frequency synchronization)

Kuramoto model with memory

Kuramoto model with memory

Motivated by viscoelastic material formulated by Volterra-type equation.
[Cho, Ha, and K., Submitted]

$$\dot{\theta}_i = \nu_i + \frac{\kappa}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i)$$



$$\dot{\theta}_i(t) = \nu_i + \frac{\kappa}{N} \sum_{j=1}^N \int_0^t a(t-s) \sin(\theta_j(s) - \theta_i(s)) ds$$

Remark. $a \geq 0$ and $\int_0^t a(s) ds < \infty$ for all $t \geq 0$

Kuramoto model with memory

[Cho, Ha, and K., Submitted]

$$m\ddot{\theta}_i + \gamma\dot{\theta}_i = \nu_i + \frac{\kappa}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i)$$

$\Downarrow$ (integrate)

$$\dot{\theta}_i(t) = \frac{\nu_i}{\gamma} + \left(\dot{\theta}_i(0) - \frac{\nu_i}{\gamma} \right) e^{-\frac{\gamma}{m}t} + \frac{\kappa}{N} \sum_{j=1}^N \int_0^t \frac{1}{m} e^{-\frac{\gamma}{m}(t-s)} \sin(\theta_j(s) - \theta_i(s)) ds$$

$\Downarrow$ ($\dot{\theta}_i(0) = \tilde{\nu}_i := \nu_i/\gamma$)

$$\dot{\theta}_i(t) = \tilde{\nu}_i + \frac{\kappa}{N} \sum_{j=1}^N \int_0^t a(t-s) \sin(\theta_j(s) - \theta_i(s)) ds$$

Kuramoto model with memory ($N = 2$)

$$\begin{cases} \dot{\theta}_1(t) = \nu_1 + \frac{\kappa}{2} \int_0^t a(t-s) \sin(\theta_2(s) - \theta_1(s)) ds, \\ \dot{\theta}_2(t) = \nu_2 + \frac{\kappa}{2} \int_0^t a(t-s) \sin(\theta_1(s) - \theta_2(s)) ds. \end{cases}$$

$$\Downarrow \quad (\theta := \theta_1 - \theta_2, \quad \nu := \nu_1 - \nu_2)$$

$$\dot{\theta}(t) = \nu - \kappa \int_0^t a(t-s) \sin(\theta(s)) ds = \nu - \kappa(a * \sin \theta)(t)$$

$$\Downarrow \quad (|\theta| \ll 1)$$

$$\dot{\theta}(t) \approx \nu - \kappa(a * \theta)(t)$$

Effect of memory ($a \equiv 1$)

Let $\hat{a}$ be Laplace's transform of θ .

$$\dot{\theta} = \nu - \kappa(a * \theta) \implies \hat{\theta}(s) = \frac{s\theta(0)}{s^2 + \kappa} + \frac{\nu}{s^2 + \kappa}$$

$$\implies \begin{cases} \theta(t) = \theta(0) \cos \sqrt{\kappa}t + \frac{\nu}{\sqrt{\kappa}} \sin \sqrt{\kappa}t, \\ \dot{\theta}(t) = -\theta(0)\sqrt{\kappa} \sin \sqrt{\kappa}t + \nu \cos \sqrt{\kappa}t. \end{cases}$$

$\implies$ No Synchronization!

Effect of memory ($a(t) = \exp(-\lambda t)$)

$$\hat{\theta}(s) = \frac{\theta^0(s + \frac{\lambda}{2})}{(s + \frac{\lambda}{2})^2 + \kappa - \frac{\lambda^2}{4}} + \frac{\frac{\lambda}{2}\theta^0}{(s + \frac{\lambda}{2})^2 + \kappa - \frac{\lambda^2}{4}} + \frac{\nu\lambda}{\kappa s} \\ + \frac{-\frac{\nu\lambda}{\kappa}(s + \frac{\lambda}{2})}{(s + \frac{\lambda}{2})^2 + \kappa - \frac{\lambda^2}{4}} + \frac{\nu(1 - \frac{\lambda^2}{2\kappa})}{(s + \frac{\lambda}{2})^2 + \kappa - \frac{\lambda^2}{4}}$$

$$\Rightarrow \left\{ \begin{array}{l} \theta(t) = \theta(0)e^{-\frac{\lambda}{2}t} \cos\left(\sqrt{\kappa - \frac{\lambda^2}{4}}t\right) + \frac{\lambda\theta(0)}{2\sqrt{\kappa - \frac{\lambda^2}{4}}}e^{-\frac{\lambda}{2}t} \sin\left(\sqrt{\kappa - \frac{\lambda^2}{4}}t\right) \\ \quad + \frac{\nu\lambda}{\kappa} - \frac{\nu\lambda}{\kappa}e^{-\frac{\lambda}{2}t} \cos\left(\sqrt{\kappa - \frac{\lambda^2}{4}}t\right) + \frac{\nu(1 - \frac{\lambda^2}{2\kappa})}{\sqrt{\kappa - \frac{\lambda^2}{4}}}e^{-\frac{\lambda}{2}t} \sin\left(\sqrt{\kappa - \frac{\lambda^2}{4}}t\right) \quad (\kappa > \lambda^2/4) \\ \\ \theta(t) = \theta^0 e^{-\frac{\lambda}{2}t} + \frac{\lambda\theta^0}{2}te^{-\frac{\lambda}{2}t} + \frac{\nu\lambda}{\kappa} - \frac{\nu\lambda}{\kappa}e^{-\frac{\lambda}{2}t} - \nu\left(1 - \frac{\lambda^2}{2\kappa}\right)te^{-\frac{\lambda}{2}t} \quad (\kappa = \lambda^2/4) \\ \\ \theta(t) = \theta^0 e^{-\frac{\lambda}{2}t} \cosh\left(t\sqrt{\frac{\lambda^2}{4} - \kappa}\right) + \frac{\lambda\theta^0}{2\sqrt{\frac{\lambda^2}{4} - \kappa}}e^{-\frac{\lambda}{2}t} \sinh\left(t\sqrt{\frac{\lambda^2}{4} - \kappa}\right) + \frac{\nu\lambda}{\kappa} \\ \quad - \frac{\nu\lambda}{\kappa}e^{-\frac{\lambda}{2}t} \cosh\left(t\sqrt{\frac{\lambda^2}{4} - \kappa}\right) + \frac{\nu(1 - \frac{\lambda^2}{2\kappa})}{\sqrt{\frac{\lambda^2}{4} - \kappa}}e^{-\frac{\lambda}{2}t} \sinh\left(t\sqrt{\frac{\lambda^2}{4} - \kappa}\right) \quad (\kappa < \lambda^2/4) \end{array} \right.$$

$\Rightarrow$ Synchronization always occurs!

Synchronization under the effect of memory

Complete Frequency Synchronization

$$\dot{\theta}_i(t) = \nu_i + \frac{\kappa}{N} \sum_{j=1}^N \int_0^t a(t-s) \sin(\theta_j(s) - \theta_i(s)) ds$$

Under what condition

$$\lim_{t \rightarrow \infty} |\dot{\theta}_i - \dot{\theta}_j| = 0 \iff \lim_{t \rightarrow \infty} \dot{\theta}_i = 0 ?$$

(Galilean invariance)

- **Case A:** a has compact support.
- **Case B:** a does not have compact support but monotonically decrease.

Frequency synchronization - Case A

Suppose a is **compactly supported** in $[0, \tau]$.

$$\dot{\theta}_i(t) = \nu_i + \frac{\kappa}{N} \sum_{j=1}^N \int_0^\tau a(s) \sin(\theta_j(t-s) - \theta_i(t-s)) ds, \quad \forall t > \tau.$$

$$\Downarrow \quad (\omega_i = \dot{\theta}_i)$$

$$\dot{\omega}_i(t) = \frac{\kappa}{N} \sum_{j=1}^N \int_0^\tau a(s) (\omega_j(t-s) - \omega_i(t-s)) \cos(\theta_j(t-s) - \theta_i(t-s)) ds$$

Claim. $\lim_{t \rightarrow \infty} |\omega_i - \omega_j| = 0, \quad \forall i, j = 1, \dots, N.$

Frequency synchronization - Case A

Let $\omega_{ij} := \omega_i - \omega_j$. Under some conditions, we have

$$\frac{d}{dt}|\omega_{ij}(t)| \leq -C_1|\omega_{ij}(t)| + C_2 \left(\max_{t-\tau \leq s \leq t} \max_{i,j} |\omega_{ij}(s)| \right),$$

for some $C_1, C_2 > 0$. If $C_2/C_1 < 1$, then **bootstrap principle** implies

$$\begin{aligned} \exists T_n \nearrow \infty \quad \text{such that} \quad T_{n+1} > T_n + \tau \\ \text{and} \quad |\omega_{ij}(t)| \lesssim \left(\frac{C_2}{C_1} \right)^n, \quad \forall t \geq T_n. \end{aligned}$$

Hence, we have $\lim_{t \rightarrow \infty} |\dot{\theta}_i - \dot{\theta}_j| = 0$ for all i and j .

Frequency synchronization - Case B

Now, we consider **monotonically decreasing** $a \in L^1(\mathbb{R}_+)$.

Let $\Theta = (\theta_1, \dots, \theta_N)$ and $\mathcal{V} = (\nu_1, \dots, \nu_N)$.

$$\dot{\theta}_i(t) = \nu_i + \frac{\kappa}{N} \sum_{j=1}^N \int_0^t a(t-s) \sin(\theta_j(s) - \theta_i(s)) ds$$

$$\Downarrow \quad g = (g_1, \dots, g_N), \quad g_i(\Theta) = \frac{\nu_i}{\|a\|_1} + \frac{\kappa}{N} \sum_{k=1}^N \sin(\theta_k - \theta_i)$$

$$\dot{\Theta} = \underbrace{\mathcal{V} \left(1 - \frac{1}{\|a\|_1} \int_0^t a(s) ds \right)}_{\text{converges to 0 as } t \rightarrow \infty} + \int_0^t a(t-s) g(\Theta(s)) ds$$

Claim. $\lim_{t \rightarrow \infty} \dot{\Theta} = 0$

Frequency synchronization - Case B

Suppose $\lim_{t \rightarrow \infty} g(\Theta(t)) = 0$. Take $T \gg 1$ such that

$$\int_t^\infty a(s)ds < \varepsilon, \quad \|g(\Theta(t))\|_\infty < \varepsilon, \quad \forall t > T.$$

For $t > 2T$, we have

$$\begin{aligned} & \left\| \int_0^t a(t-s)g(\Theta(s))ds \right\|_\infty \\ & \leq \int_0^T a(t-s)\|g(\Theta(s))\|_\infty ds + \int_T^t a(t-s)\|g(\Theta(s))\|_\infty ds \\ & \leq \|g\|_\infty \int_T^\infty a(s)ds + \varepsilon\|a\|_1 \leq \varepsilon(\|g\|_\infty + \|a\|_1). \end{aligned}$$

Frequency synchronization - Case B

$$\dot{\Theta} = \nu \left(1 - \frac{1}{\|a\|_1} \int_0^t a(s) ds \right) + \int_0^t a(t-s) g(\Theta(s)) ds$$

Energy function. $\mathcal{E}(t) = E(t) + R(t) + V(t)$ where

$$E(t) := \frac{a(t)}{2} \left\| \int_0^t g(\Theta(s)) ds \right\|^2 - \frac{1}{2} \int_0^t a'(t-s) \left\| \int_s^t g(\Theta(u)) du \right\|^2 ds,$$

$$R(t) := - \sum_{i=1}^N \frac{\nu_i \theta_i(t)}{\|a\|_1} - \frac{\kappa}{2N} \sum_{i,k=1}^N \cos(\theta_k(t) - \theta_i(t)),$$

$$V(t) := \int_0^t \langle \nu, g(\Theta(s)) \rangle \left(1 - \frac{1}{\|a\|_1} \int_0^s a(u) du \right) ds.$$

Remark. $|R| \lesssim \|\Theta\|$ and $|V| \lesssim \int_0^\infty sa(s) ds$

Frequency synchronization - Case B

$$\mathcal{E}'(t) = \frac{a'(t)}{2} \left\| \int_0^t g(\Theta(s)) ds \right\|^2 - \frac{1}{2} \int_0^t a''(t-s) \left\| \int_s^t g(\Theta(u)) du \right\|^2 ds$$

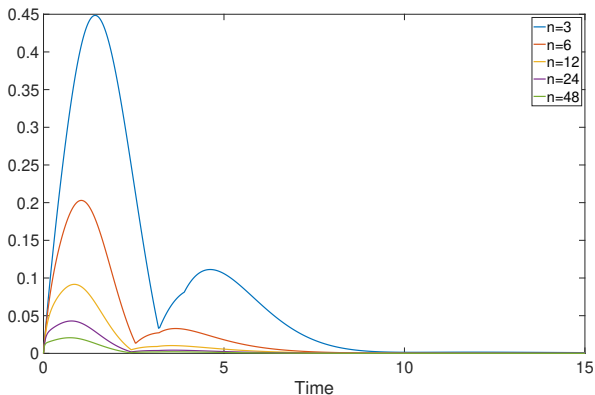
$\implies a' < 0$ and $a'' > 0$ implies monotonically decreasing energy

$\implies$ (under some conditions) $\mathcal{E}(t)$ has uniform lower bound

$\implies$ (under some conditions) $\lim_{t \rightarrow \infty} g(\Theta(t)) = 0$

Numerical results

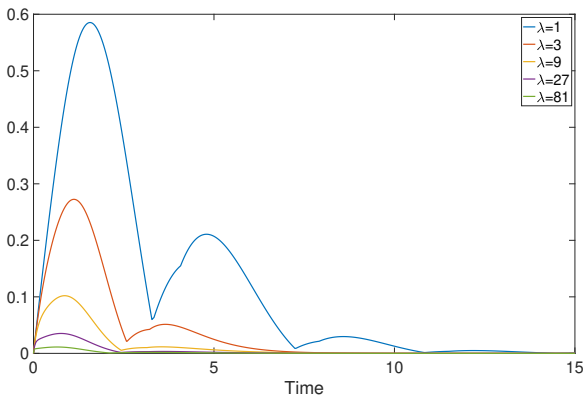
$$\dot{\phi}_i = \frac{1}{N} \sum_{k=1}^N \sin(\phi_k - \phi_i) \quad \& \quad \dot{\theta}_i = \frac{1}{N} \sum_{k=1}^N \int_0^t \frac{n-1}{(1+t-s)^n} \sin(\theta_k(s) - \theta_i(s)) ds$$



$$\max_i |\phi_i(t) - \theta_i(t)|$$

Numerical results

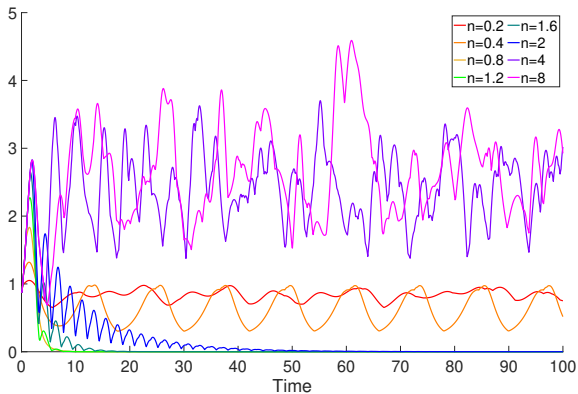
$$\dot{\phi}_i = \frac{1}{N} \sum_{k=1}^N \sin(\phi_k - \phi_i) \quad \& \quad \dot{\theta}_i = \frac{1}{N} \sum_{k=1}^N \int_0^t \lambda e^{-\lambda(t-s)} \sin(\theta_k(s) - \theta_i(s)) ds$$



$$\max_i |\phi_i(t) - \theta_i(t)|$$

Numerical results

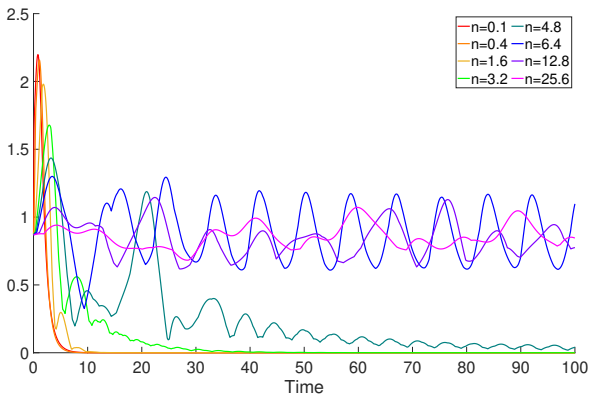
$$a(t) = \begin{cases} 1, & t \in [0, n], \\ 0, & t > n, \end{cases}$$



$$\max_{i,j} |\dot{\theta}_i(t) - \dot{\theta}_j(t)|$$

Numerical results

$$a(t) = \begin{cases} 1/n, & t \in [0, n], \\ 0, & t > n, \end{cases}$$



$$\max_{i,j} |\dot{\theta}_i(t) - \dot{\theta}_j(t)|$$

Reference

-  H. Cho, S.-Y. Ha, and M. Kang, *Phase-coupled models for synchronization with nonlocal temporal interactions*. Preprint.
-  Ha, S.-Y. and Ryoo, S. W.: *Asymptotic Phase-Locking Dynamics and Critical Coupling Strength for the Kuramoto Model*. Commun. Math. Phys. **377** (2020), 811–857.
-  Y. Kuramoto, *International symposium on mathematical problems in theoretical physics*. Lecture notes in Physics **30** (1975), 420.

Thank you and Happy Birthday!